



Final Project — EAI 6010 Week 6

College of Professional Studies, Northeastern University

EAI 6010 Applications of Artificial Intelligence

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Introduction

The Porto Seguro Safe Driver dataset presents a challenging binary classification problem: predicting whether an insurance customer will file a claim in the next year. The dataset is large and highly anonymized, containing over 595,000 observations with mixed feature types, including continuous variables, binary indicators, and categorical features. In addition, missing values are encoded as -1 rather than standard NaN, which requires careful preprocessing before modeling. Another major challenge is the severe class imbalance, as fewer than 5% of observations correspond to actual insurance claims. Under this imbalance, accuracy alone becomes a misleading evaluation metric, making it necessary to rely on measures such as AUC. In this project, we first focus on building a robust preprocessing pipeline that cleans missing values, encodes categorical variables, and standardizes numerical features. Using the processed dataset, we evaluate simple and interpretable baseline models, including logistic regression and a decision tree, to establish reference performance and better understand the structure of the problem. These baselines motivate the use of more expressive models, which are explored in the later sections of the report.

Dataset Description & Problem Setup

This project uses the Porto Seguro Safe Driver Prediction dataset from Kaggle, which focuses on estimating the risk that an auto insurance policy holder will file a claim in the future. The training dataset contains 595,212 observations, with each row representing an individual insurance policy. The response variable is binary, indicating whether a claim occurred (target =

1) or not (target = 0). An identifier column is provided in the raw data but is removed prior to analysis, as it carries no predictive information.

The feature set is fully anonymized and consists of 59 variables capturing different aspects of the policy holder and the insured vehicle. These variables include continuous numerical features, binary indicators, and categorical attributes with multiple levels. Feature names follow a structured convention such as `ps_ind_*`, `ps_reg_*`, `ps_car_*`, and `ps_calc_*`, although their precise semantic meaning is intentionally hidden. In addition, missing values are encoded using a sentinel value (-1) rather than standard missing indicators, requiring careful preprocessing before model training.

A key challenge of this dataset is the severe class imbalance. Only a small fraction of the observations correspond to policies with an actual claim, while the vast majority belong to the non-claim class. As a result, naive classifiers can achieve deceptively high accuracy by always predicting the majority class. This makes it necessary to frame the problem as an imbalanced classification task and to rely on evaluation metrics such as the area under the ROC curve (AUC), which better reflect a model's ability to discriminate between classes.

Overall, the prediction task combines several practical difficulties commonly encountered in real-world insurance analytics, including anonymized high-dimensional tabular data, mixed feature types, non-standard missing value representations, and strong class imbalance. These characteristics motivate a modeling pipeline that begins with thorough preprocessing and baseline models, and then extends to more expressive approaches explored later in the project.

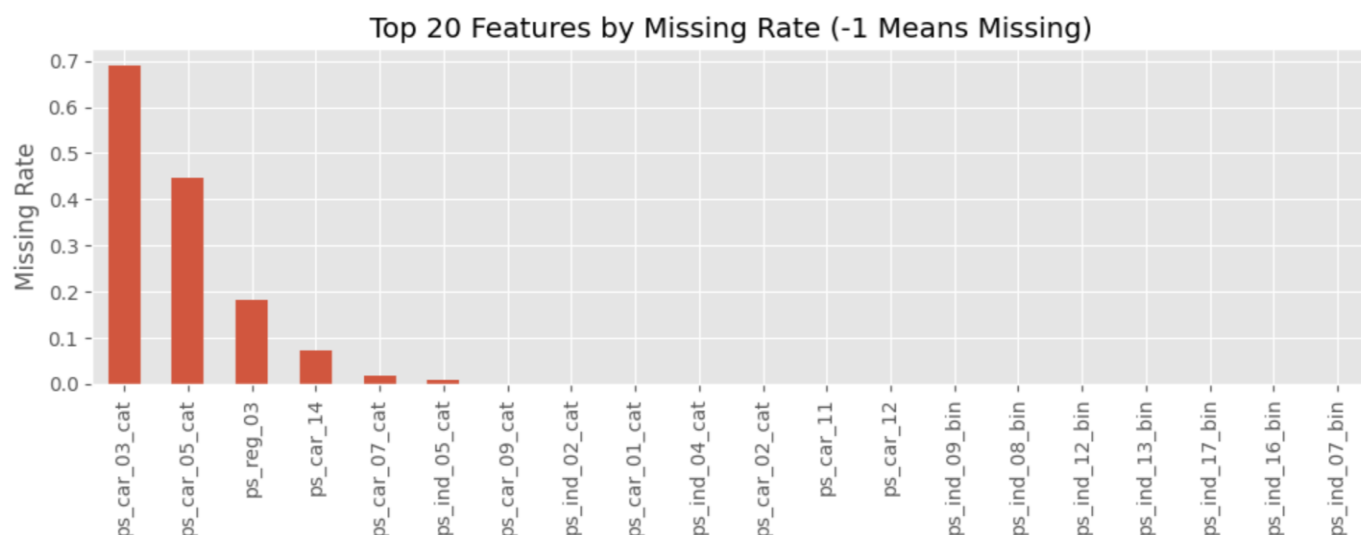
Data Exploration & Preprocessing

We use the Porto Seguro Safe Driver dataset, which contains 595,212 observations with 59 original features, excluding the ID and target variable. The target is a binary indicator of whether an insurance claim occurred. Before modeling, it is necessary to carefully preprocess the data due to heterogeneous feature types and a substantial amount of missing values.

The features are first divided into three groups based on their naming conventions and data characteristics: continuous features, binary features (ending with `_bin`), and categorical features (ending with `_cat`). This separation allows different preprocessing strategies to be applied in a controlled and interpretable way.

Missing Value Analysis

In this dataset, missing values are encoded as -1. To understand the severity and distribution of missingness, we compute the feature-wise missing rate across all predictors. Several features exhibit non-trivial levels of missing data, with `ps_car_03_cat` showing an especially high missing rate of nearly 70%. Given the extreme level of missingness, `ps_car_03_cat` is dropped from further analysis, as such features are unlikely to provide reliable predictive signal and may instead introduce noise or instability into downstream models.



Continuous Feature Processing

For continuous variables, the value -1 is treated as missing and first replaced with NaN. Missing values are then imputed using the median of each feature. Median imputation is chosen because it is robust to skewed distributions and extreme values, which are common in insurance-related data. After imputation, all continuous features are standardized using z-score normalization. Scaling is particularly important for linear models and neural networks, ensuring that features with larger numeric ranges do not dominate the learning process.

Binary Feature Processing

Binary features naturally take values of 0 or 1, representing yes/no conditions. Although missing values are rare in these variables, any occurrence of -1 is treated as missing. Since binary variables have only two valid states, missing values are imputed with the mode of the observed values in each feature. This approach preserves the original semantic meaning of binary indicators and avoids introducing artificial numeric values into the model.

Categorical Feature Processing

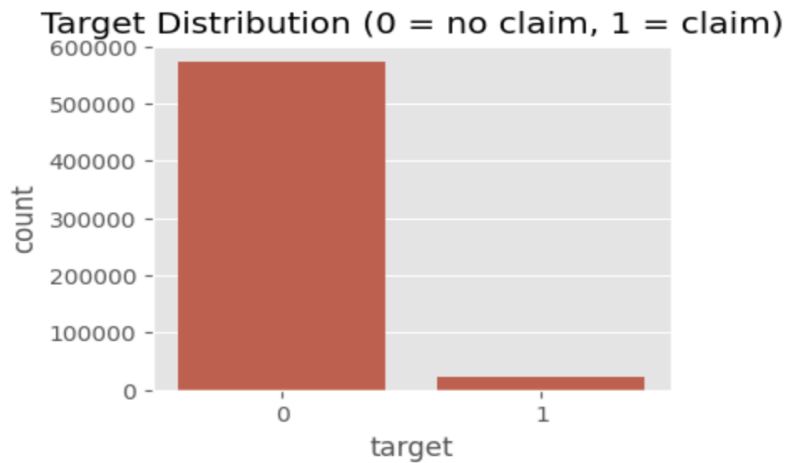
Categorical features are handled differently due to their discrete and unordered nature. After removing extremely high-missing features, the remaining categorical variables retain -1 as an explicit category representing “unknown” or “missing,” rather than being imputed. All categorical features are then transformed using one-hot encoding. This avoids imposing any ordinal relationship between category IDs and allows models to learn independent effects for each category. After preprocessing, the categorical feature matrix expands from 13 columns to 181 one-hot encoded columns.

The final feature matrix is obtained by concatenating the scaled continuous features, cleaned binary features, and one-hot encoded categorical features column-wise. This results in a design matrix with 224 features, which serves as the input for all subsequent models. At this stage, the dataset is fully numeric, free of artificial missing codes, and appropriately scaled and encoded for machine learning algorithms.

Final prepared feature matrix shape: (595212, 224)											
	ps_ind_01	ps_ind_03	ps_ind_14	ps_ind_15	ps_reg_01	ps_reg_02	ps_reg_03	ps_car_11	ps_car_12	ps_car_13	...
0	0.050218	0.213594	-0.097621	1.043440	0.309442	-0.591654	-0.506132	-0.415746	0.343963	0.313526	...
1	-0.453868	0.954362	-0.097621	-1.212598	0.657096	-0.096928	-0.353536	0.785466	-1.092966	-0.865800	...
2	1.562477	1.695130	-0.097621	1.325445	-2.124135	-1.086380	-0.240749	-1.616957	-1.092966	-0.764418	...
3	-0.957955	-0.897559	-0.097621	0.197425	1.004750	-0.591654	-0.941992	-1.616957	-0.099167	-1.203608	...
4	-0.957955	-1.638327	-0.097621	0.479430	0.309442	0.397799	-0.116155	0.785466	-1.095679	-1.101721	...
5 rows x 224 columns											

Target Distribution and Class Imbalance

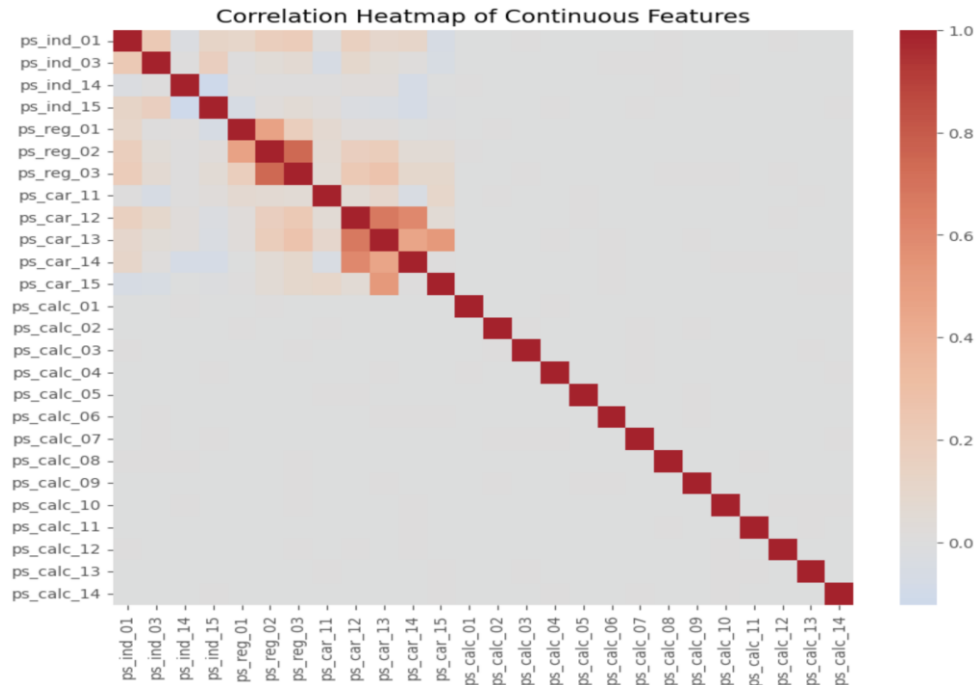
The target variable is highly imbalanced. Approximately 96.4% of the observations correspond to no insurance claim (target = 0), while only about 3.6% represent a claim (target = 1). This imbalance has important implications for model evaluation and metric selection.



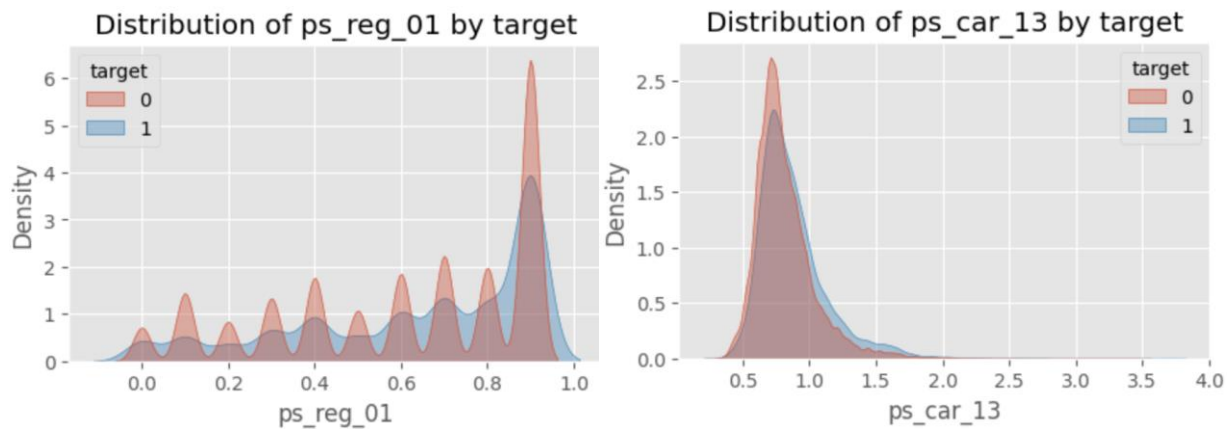
Because a naive classifier that predicts only the negative class would already achieve very high accuracy, accuracy alone is not a meaningful metric for this task. Metrics such as AUC, recall, and class-weighted loss functions are therefore more appropriate.

Correlation Analysis and Feature Distributions

To further explore the structure of the data, we examine correlations among continuous features. The correlation heatmap reveals that several continuous variables form clusters with moderate to high correlations, indicating potential redundancy in the feature space.



To better understand class separation, we visualize the distributions of representative continuous features (ps_reg_01 and ps_car_13) stratified by the target variable. While some overlap remains, subtle distributional shifts between classes suggest that these features may carry predictive signal.



Model Development

Baseline Models

We spent some time to test a few models after the completion of the preprocessing pipeline to get an intuition on how various approaches would work on the Porto Seguro dataset. As the initial models, we chose to test logistic regression and a decision tree. The motivation for these choices is that these models are simple and interpretable, which made them a logical first step. However, both models only achieved a validation AUC of around 0.61 and the accuracies were either low (or very high, depending on the threshold) as a result of class imbalance. In this dataset, approximately 95% of samples are of class zero, which means a zero-accuracy model has a baseline accuracy of about 96 percent. This first experiment demonstrated to me why accuracy is not an appropriate metric to use for the evaluation of this problem, and why AUC is a better metric for imbalanced datasets.

Advanced Models

The poor results from these baseline models also illuminated the data's insufficiency for models with traditional inductive biases. As mentioned above, logistic regression uses linear assumptions. Although we balanced the classes, the shallow decision tree still could not find meaningful nonlinear interactions. To further investigate what types of models were likely to work better, we reviewed a recent survey on the use of deep learning for tabular data (Borisov et al., 2022). The authors state that tabular data presents a challenge for classical models because “many classical machine learning models fail on high-dimensional and anonymized features, unless the features are hand-engineered to model the hidden interactions” (Borisov et al., 2022). In other words, classical models are not suitable for tabular data with many features whose

semantics are hidden. This paper’s description resonates with the structure of our Porto Seguro dataset in the sense that no manual feature engineering can be performed as the feature semantics are unknown.

Motivated by this reason, we decided to use a neural network as a way to extract some information about this dataset. According to the survey above, deep learning works on tabular data if there is complex or heterogeneous structure (Sarker, 2021). A neural network works by learning layer-wise nonlinear transformations, which is more appropriate for learning patterns from a dataset where the meaning of each feature is unknown. In this way, we were convinced to train a simple MLP as the next step in the pipeline.

```
*** ImprovedMLP(  
    (net): Sequential(  
      (0): Linear(in_features=224, out_features=128, bias=True)  
      (1): ReLU()  
      (2): Dropout(p=0.3, inplace=False)  
      (3): Linear(in_features=128, out_features=64, bias=True)  
      (4): ReLU()  
      (5): Dropout(p=0.3, inplace=False)  
      (6): Linear(in_features=64, out_features=1, bias=True)  
    )  
  )  
)
```

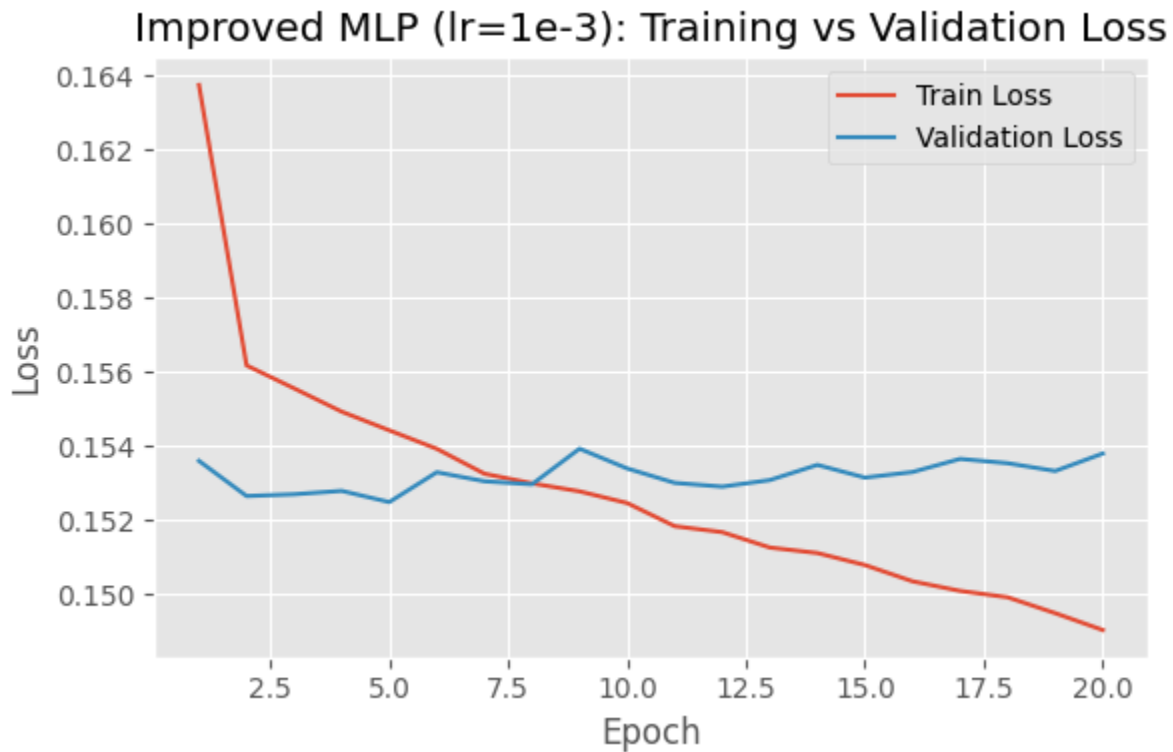
Model Evaluation & Results

Model Performance

Baseline MLP’s AUC was about 0.63 to 0.64. Compared to the results from traditional models, the result is improved, which means the network is able to learn more structure of the data.

However, after increasing the architecture to two hidden layers, the performance failed to improve. We later realized that the training curves were not stable, and the network’s weights did

not converge during training.



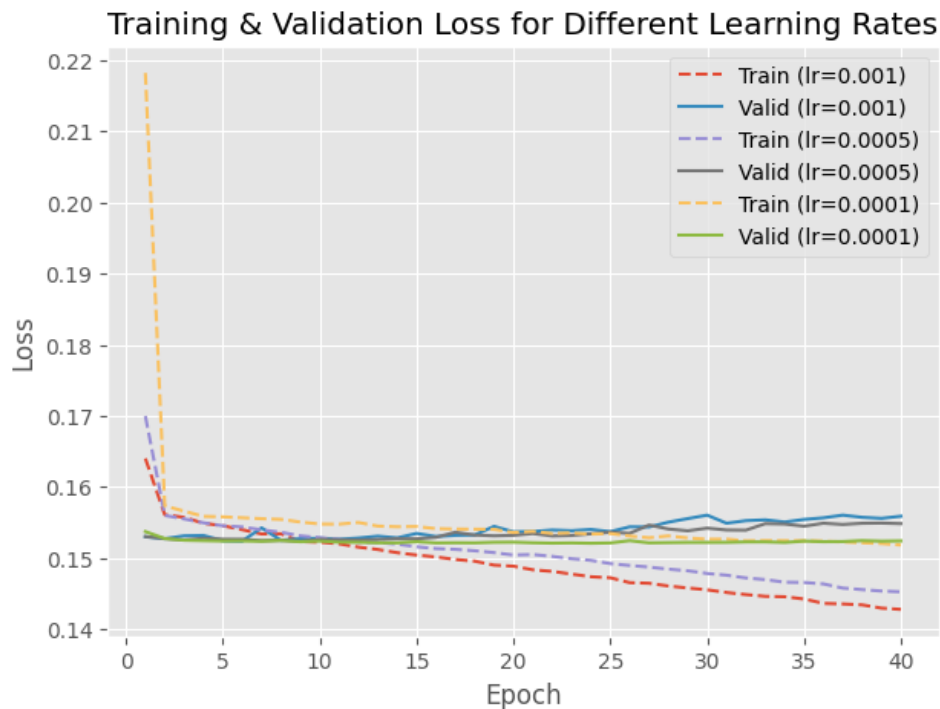
We

found that the shallow neural network is more robust to changes in hyperparameters, and that deeper models are more sensitive to optimization settings such as the learning rate. Deep learning optimization research has long found that the learning rate dominates the convergence behavior and stability of neural network models (Smith, 2018). This lesson pushed us to run an experiment on learning rate with values of $1e-3$, $5e-4$ and $1e-4$.

Advanced Model Performance

Adding early stopping to the training loop and training for 40 epochs, we suppose that the deeper MLP would eventually be halted early as it started overfitting. This was never the case, early stopping was not triggered for any learning rate. While this is a less than useful result, it also served as a confirmation of an important point: The model was not overfitting, it was just not

improving. The training loss kept gradually decreasing, while the validation loss fluctuated in a tight band and the validation AUC was virtually flat. The results indicate that the constraint here is not training the model too long, but the model's capacity and the sensitivity to learning rate. In this light, out of the three tested values, $1e-4$ gave the most regular validation curve, confirming it as the best configuration, even if the performance gains were marginal.



Limitations & Future Work

There are a number of possible extensions to this work that could be explored further. It might be worth experimenting with different strategies to account for the strong class imbalance in the dataset, in the hope that more effective use of the minority samples can help improve the discrimination of positive examples. Other hyperparameters could be

tuned systematically, for example by adding additional layers to the MLP or more systematically searching for learning rate schedules and regularization strengths. Finally, a more thorough evaluation using cross-validation could provide a more stable view of the model performance and reduce the impact of variance from the single train-validation split. These ideas represent some of the most obvious ways to continue the work if time permits.

Conclusion

In this project, a number of different modeling approaches were applied to the Porto Seguro insurance dataset and their performance compared using a consistent evaluation framework. The baseline methods provided a reference point and highlighted some of the issues in the dataset itself, including the imbalance and anonymized features. Switching to a neural network provided a better understanding of the behavior of nonlinear models in this setting, as well as showing the impact of choices like the learning rate. A series of experiments led to the choice of a smaller learning rate as the most stable and reliable option, although the performance gains were modest. Overall, the results illustrate that model behavior is affected by many different factors and that a careful experimental process is needed to select an appropriate approach for this type of prediction task.

Reference

Borisov, V., Leemann, T., Seßler, K., Haug, J., Pawelczyk, M., & Kasneci, G. (2022). *Deep neural networks and tabular data: A survey*. ACM.
https://www.researchgate.net/publication/366552986_Deep_Neural_Networks_and_Tabular_Data_A_Survey

Sarker, I. H. (2021). *Deep learning: A comprehensive overview on techniques, taxonomy, applications and research directions*. SN Computer Science. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8372231/>

A disciplined approach to neural network hyper-parameters: Part 1 learning rate, batch size, momentum, and weight decay. <https://arxiv.org/abs/1803.09820>